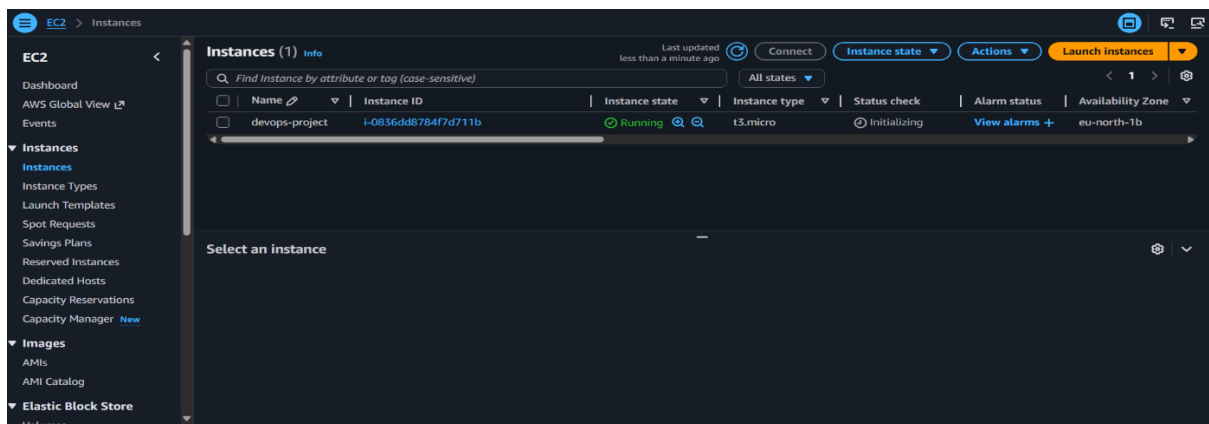
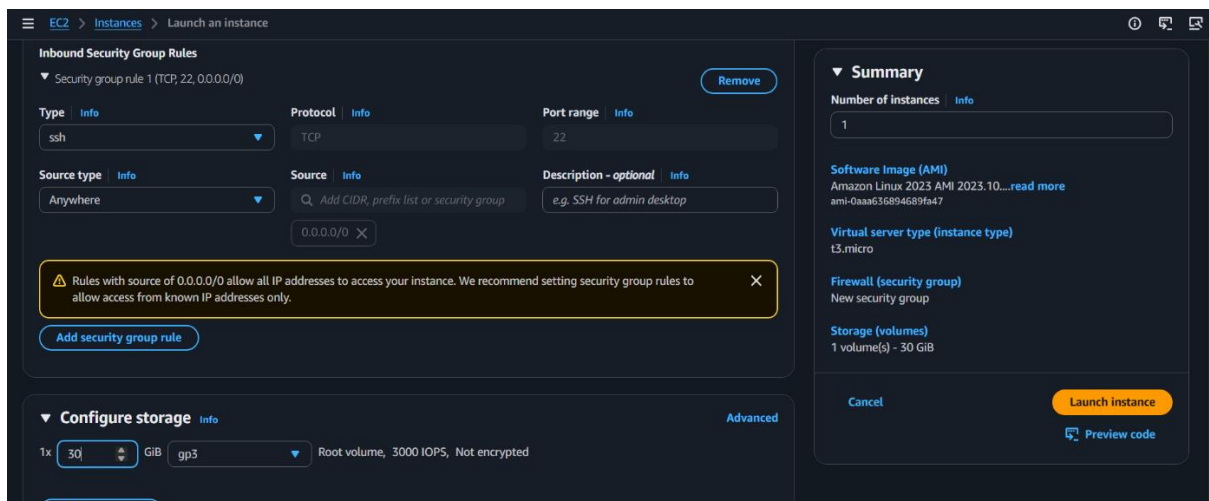
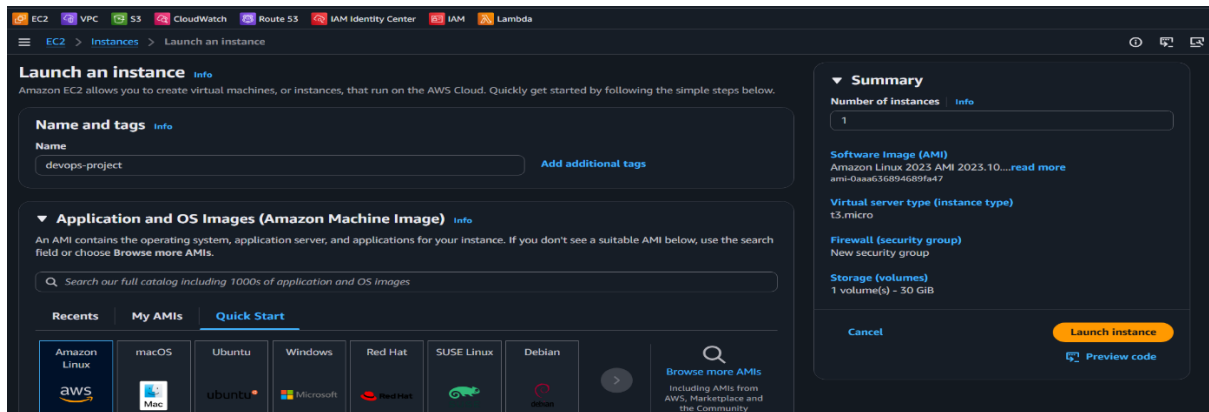


DEVOPS INTERNSHIP

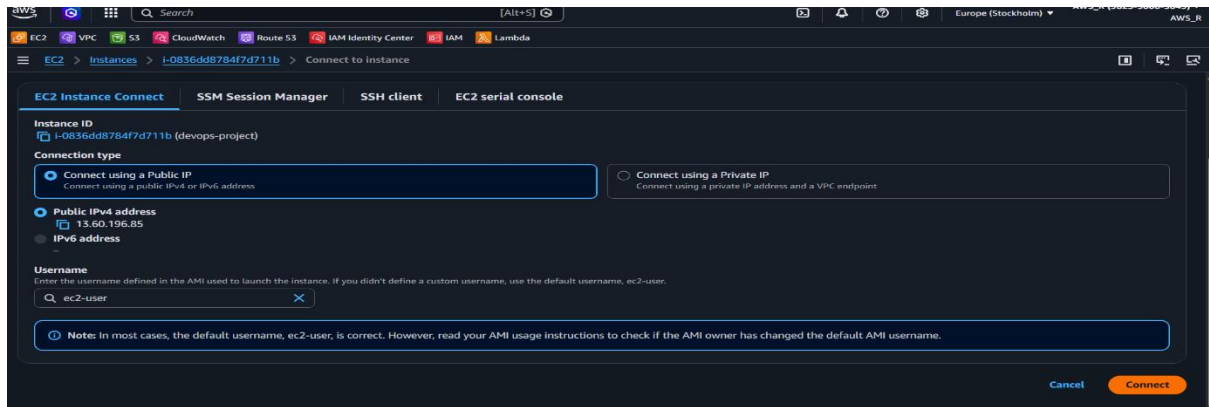
Task-1

Git Configuration=>

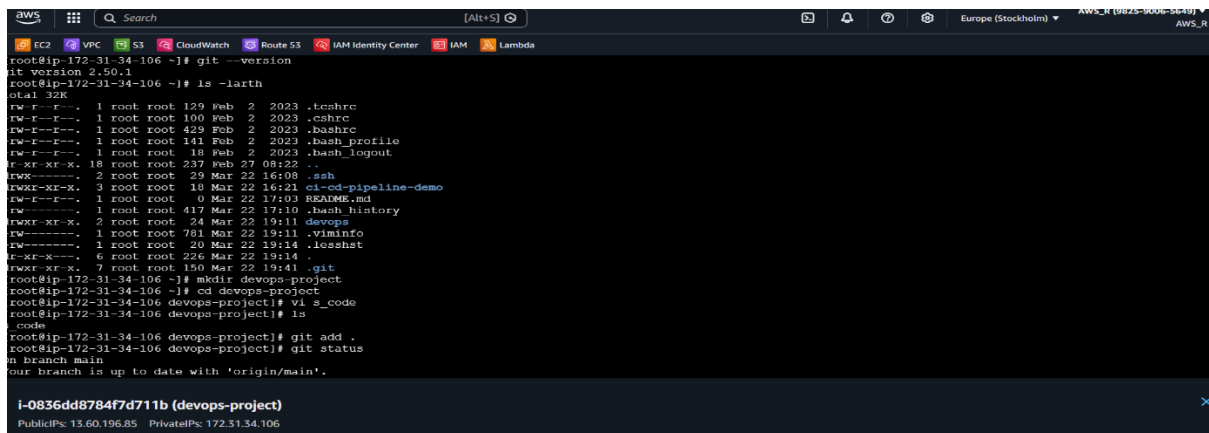
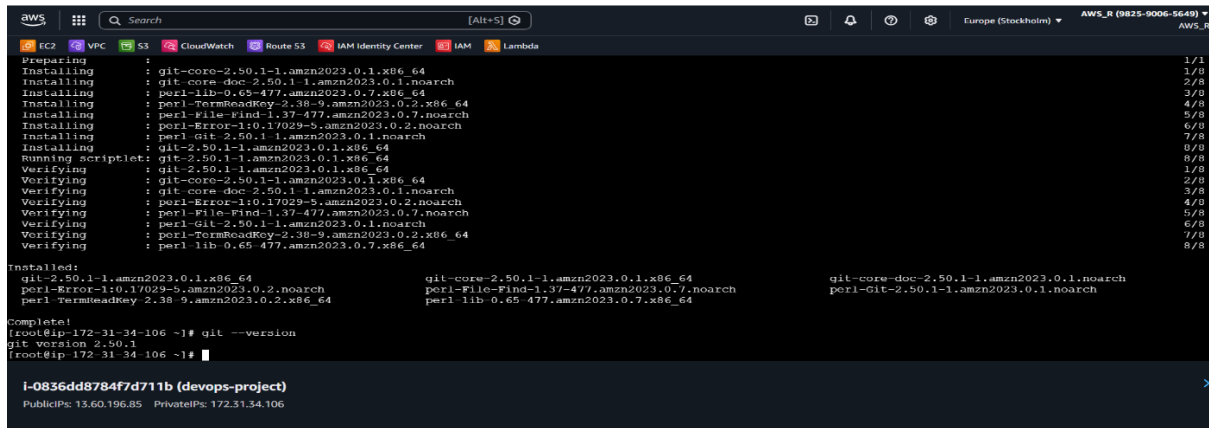
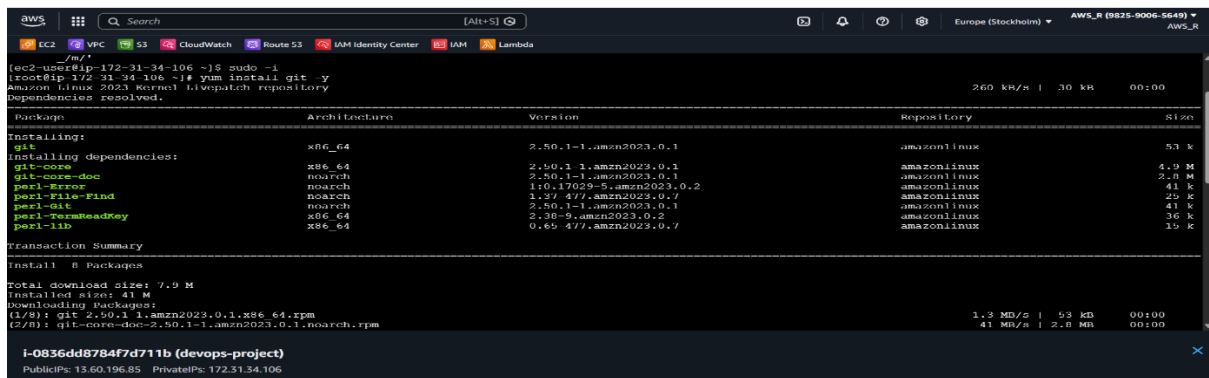
➔ launch the ec2 server and configure the server



➔ connect the instance



➔ Then install the git



```
1 file changed, 9 insertions(+)
create mode 100644 devops-project/s_code
root@ip-172-31-34-106 devops-project# git log
commit 7690fe1b21f047908ce93a2f2c056fe20425f70 (HEAD -> main)
author: root <root@ip-172-31-34-106.eu-north-1.compute.internal>
date:   Sun Mar 22 19:51:04 2026 +0000

    this is a devops-Intern Project Code

commit fdf675d329363a39cb2eb9e2801d9e5411052b4 (origin/main)
author: root <root@ip-172-31-34-106.eu-north-1.compute.internal>
date:   Sun Mar 22 19:14:23 2026 +0000

    This is the dev sourcecode

commit ee145077c1e38b019353d77de0a951321b0e287
author: root <root@ip-172-31-34-106.eu-north-1.compute.internal>
date:   Sun Mar 22 17:03:40 2026 +0000

    first commit
root@ip-172-31-34-106 devops-project# git remote add origin https://github.com/rajeswaril14v/ci-cd-pipeline-demo.git
git branch -M main
git push -u origin main
error: remote origin already exists.
username for 'https://github.com': rajeswaril14v
password for 'https://rajeswaril14v@github.com':
nummerating objects: 5, done.

i-0836dd8784f7d711b (devops-project)
PublicIPs: 13.60.196.85 PrivateIPs: 172.31.34.106
```

```
eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressfamily=ipv4&connType=standard&instanceId=i-0836dd8784f7d711b&osUser=ec2-user&region=...
AWS
EC2 VPC S3 CloudWatch Route 53 IAM Identity Center IAM Lambda
root@ip-172-31-34-106 ~# mkdir devops-project
root@ip-172-31-34-106 ~# cd devops-project
root@ip-172-31-34-106 devops-project# vi s_code
root@ip-172-31-34-106 devops-project# ls
s_code
root@ip-172-31-34-106 devops-project# git add .
root@ip-172-31-34-106 devops-project# git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   s_code

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        ../.bash_history
        ../.bash_logout
        ../.bash_profile
        ../.bashrc
        ../.cshrc
        ../.lessdist
        ../.ssh/
        ../.viminfo
        ../ci-cd-pipeline-demo/

i-0836dd8784f7d711b (devops-project)
PublicIPs: 13.60.196.85 PrivateIPs: 172.31.34.106
```

New repository

Type to search

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

1 General

Owner *

rajeswaril14v

 / Repository name *

ci-cd-pipeline-demo

Great repository names are short and memorable. How about [special-octo-palm-tree?](#)

Description

DevOps-project

14 / 300 characters

2 Configuration

Choose visibility *

Public

Choose who can see and commit to this repository

Add README

off

READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore

No .gitignore

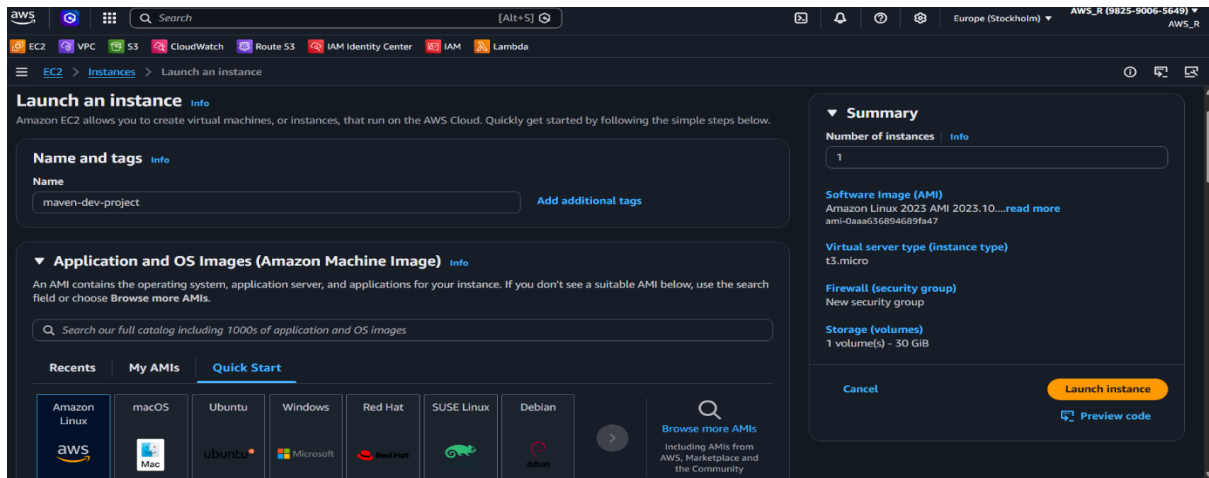
.gitignore tells git which files not to track. [About ignoring files](#)

➔ our GitHub repository set up is done

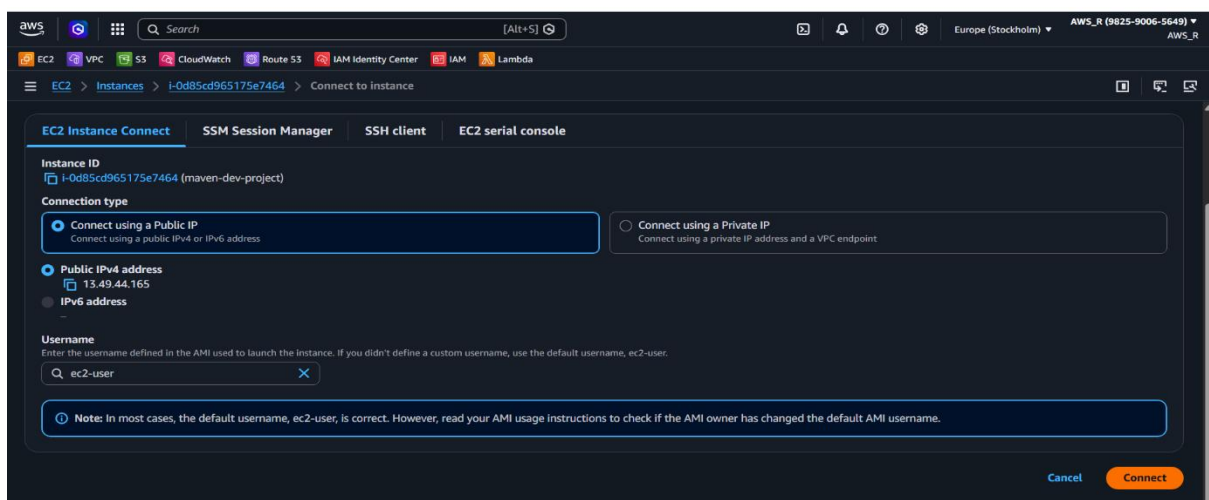
TASK-2

Maven =>

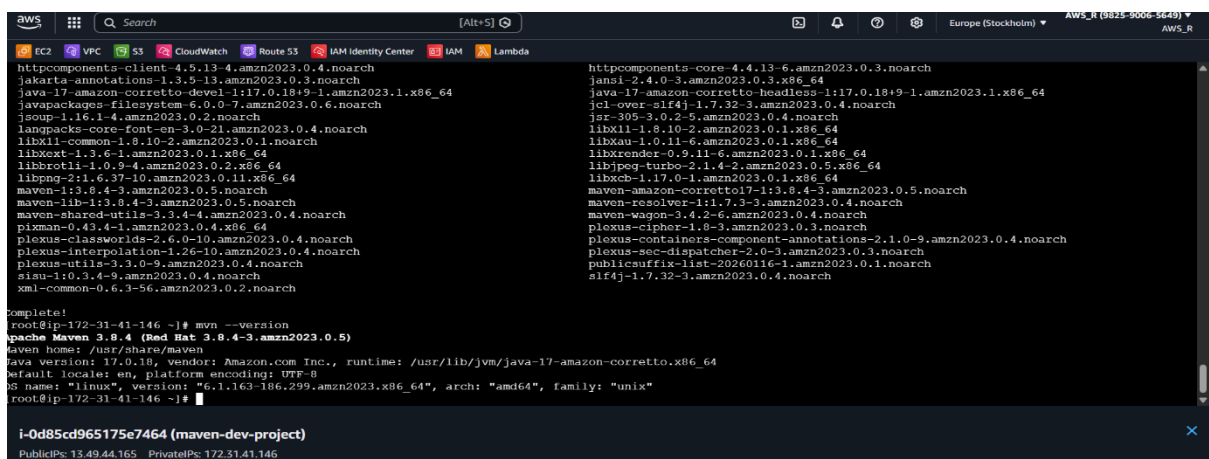
➔ create aws server name as -> Maven-dev-project

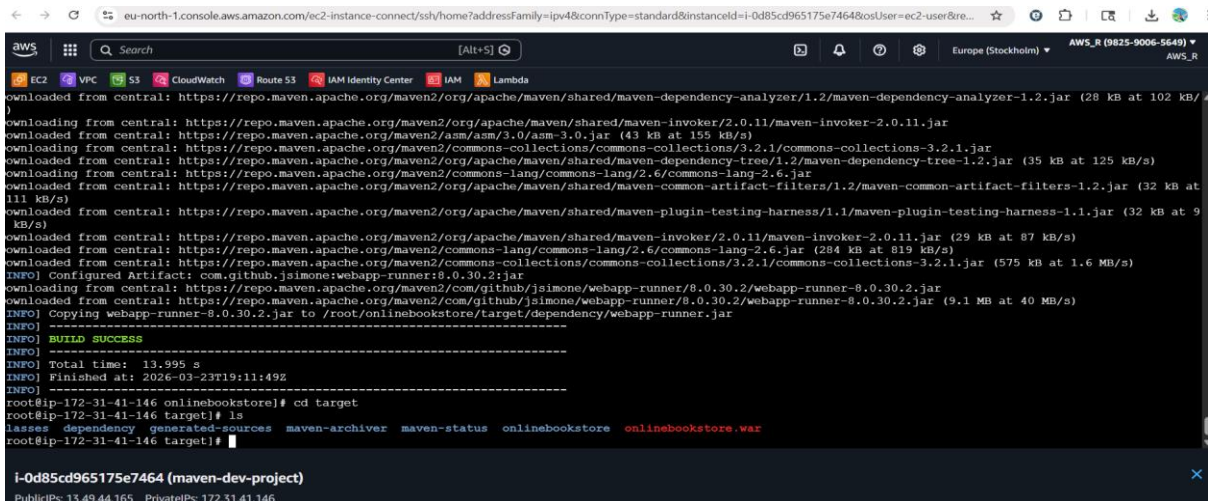
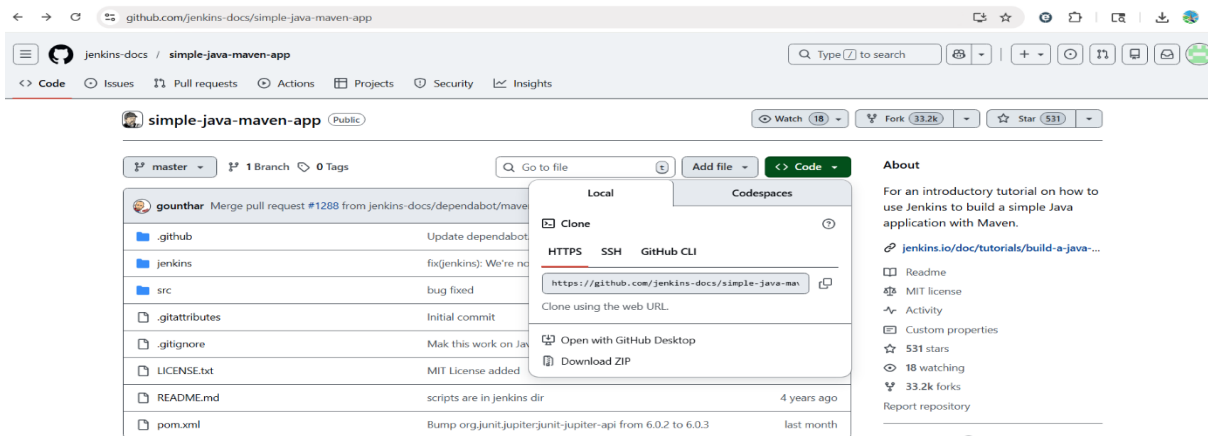
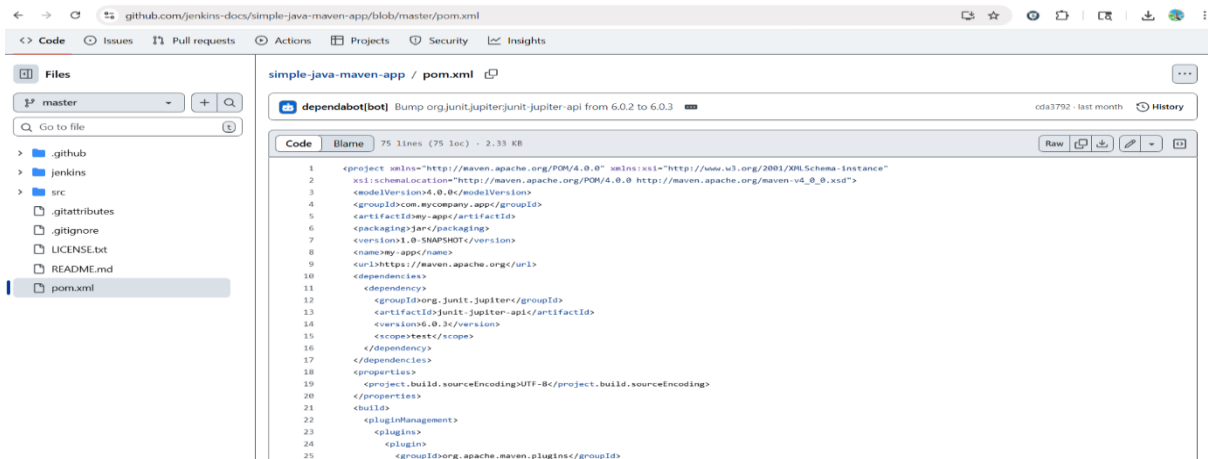


➔ And connect the server



➔ in terminal run the -> yum install maven -> mvn --version



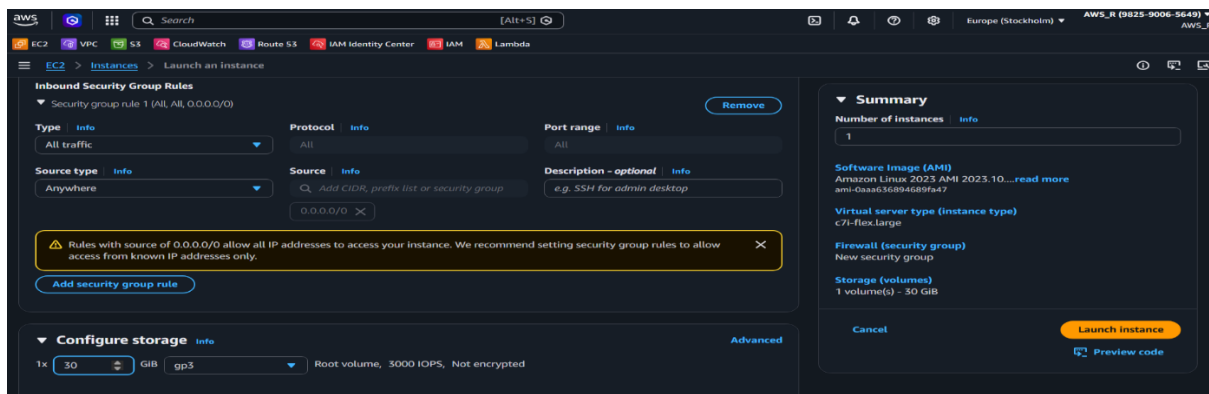
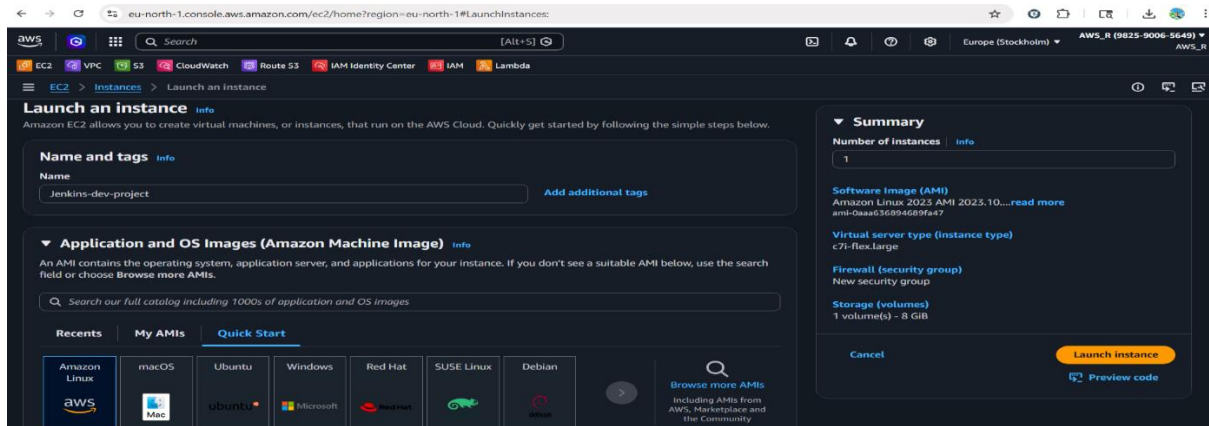


➔ maven will be updated

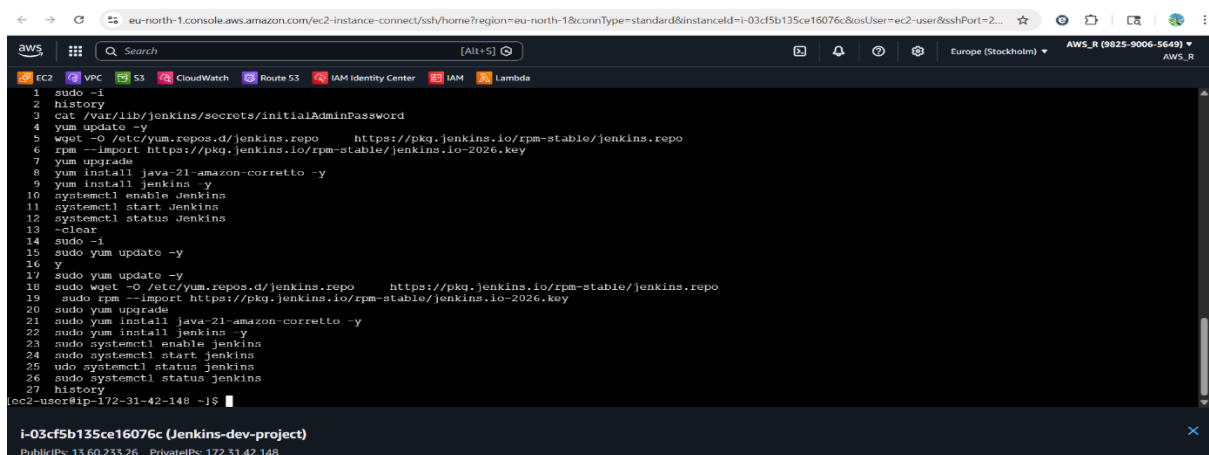
TASK-3

Jenkins => Jenkins installation and configuration =>

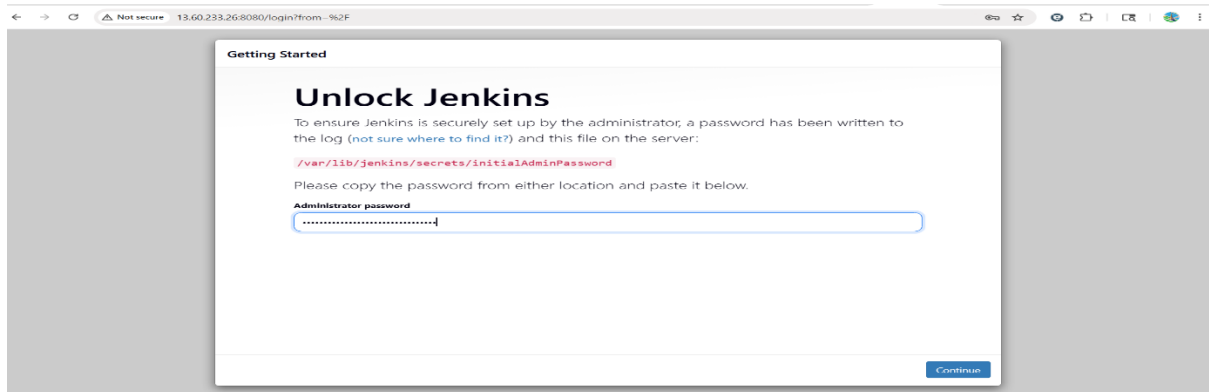
➔ Launch the Instance->Jenkins-dev-project



➔ Then Run the following commands



➔ Unlock Jenkins



The screenshot shows the 'Getting Started' page of Jenkins. The main heading is 'Unlock Jenkins'. Below it, a message states: 'To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server: /var/lib/jenkins/secrets/initialAdminPassword'. It then asks the user to 'Please copy the password from either location and paste it below.' There is a text input field labeled 'Administrator password' with a masked password '*****'. A 'Continue' button is at the bottom right.

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

`/var/lib/jenkins/secrets/initialAdminPassword`

Please copy the password from either location and paste it below.

Administrator password

Continue

➔ Customize Jenkins



The screenshot shows the 'Getting Started' page of Jenkins with the heading 'Customize Jenkins'. It says 'Plugins extend Jenkins with additional features to support many different needs.' There are two main sections: 'Install suggested plugins' (with the text 'Install plugins the Jenkins community finds most useful.') and 'Select plugins to install' (with the text 'Select and install plugins most suitable for your needs.').

Getting Started

Customize Jenkins

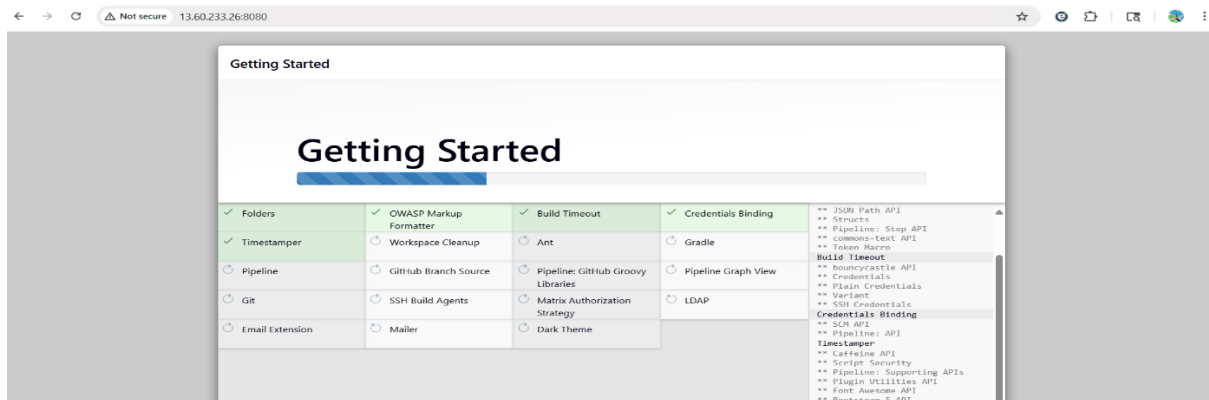
Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.



The screenshot shows the 'Getting Started' page of Jenkins with the heading 'Getting Started'. It displays a table of plugins that can be selected for installation. The table has columns for 'Suggested', 'Selected', and 'Available'. The 'Available' column lists various plugins like 'Jenkins Path API', 'Structs', 'Pipeline: Step API', etc. The 'Selected' column shows plugins like 'Folders', 'Timestampers', 'Pipeline', etc. The 'Suggested' column shows plugins like 'OWASP Markup Formatter', 'Workspace Cleanup', etc.

Getting Started

Suggested	Selected	Available
✓ Folders	✓ OWASP Markup Formatter	✓ Credentials Binding
✓ Timestampers	✓ Workspace Cleanup	✓ Ant
✓ Pipeline	✓ Git	✓ Pipeline: GitHub Groovy Libraries
✓ Git	✓ SSH Build Agents	✓ Matrix Authorization Strategy
✓ Email Extension	✓ Mailer	✓ Dark Theme

Build Timeout

- ✓ JSP Path API
- ✓ Structs
- ✓ Pipeline: Step API
- ✓ Commons-Text API
- ✓ Token Macro
- ✓ Build Timeout
- ✓ HttpClient API
- ✓ Credentials
- ✓ Plain Credentials
- ✓ Variant
- ✓ SSH Credentials
- ✓ Credentials Binding
- ✓ SSH API
- ✓ Pipeline: API
- ✓ Timestampers
- ✓ Caffeine API
- ✓ Script Security
- ✓ Pipeline: Supporting APIs
- ✓ Plugin Utilities API
- ✓ Font Awesome API
- ✓ Bootstrap 5 API



The screenshot shows the 'Getting Started' page of Jenkins with the heading 'Create First Admin User'. It contains a form with fields for 'Username', 'Password', 'Confirm password', 'Full name', and 'E-mail address'. The 'Username' field has 'admin' entered. The 'E-mail address' field has 'rajeshwari114v@gmail.com' entered. At the bottom, there are two buttons: 'Skip and continue as admin' and 'Save and Continue'.

Getting Started

Create First Admin User

Username

admin

Password

Confirm password

Full name

admin

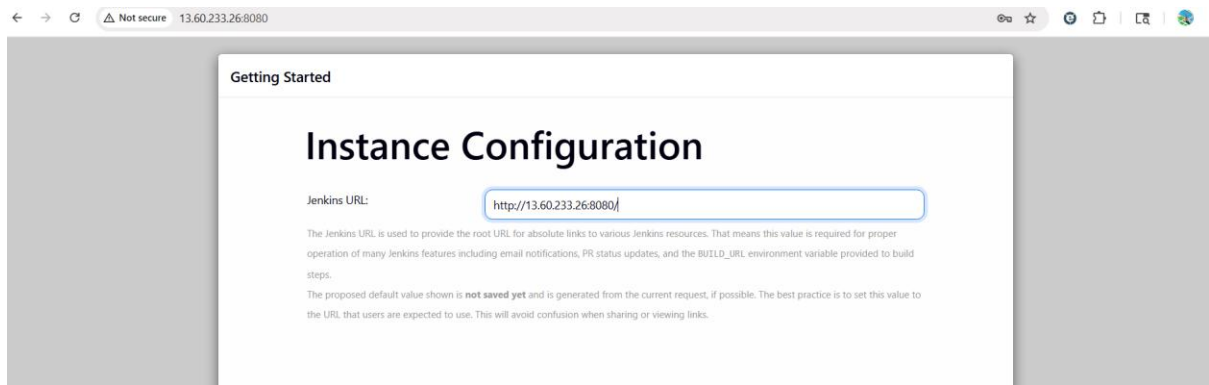
E-mail address

rajeshwari114v@gmail.com

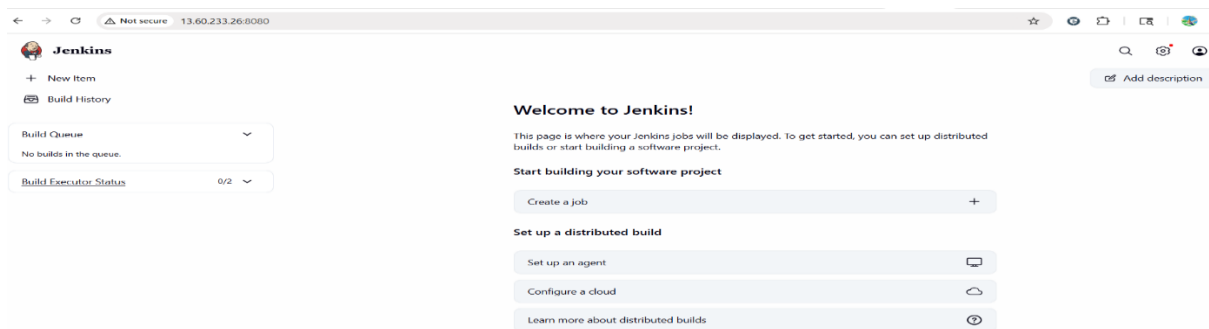
Jenkins 2.511.3

Skip and continue as admin

Save and Continue



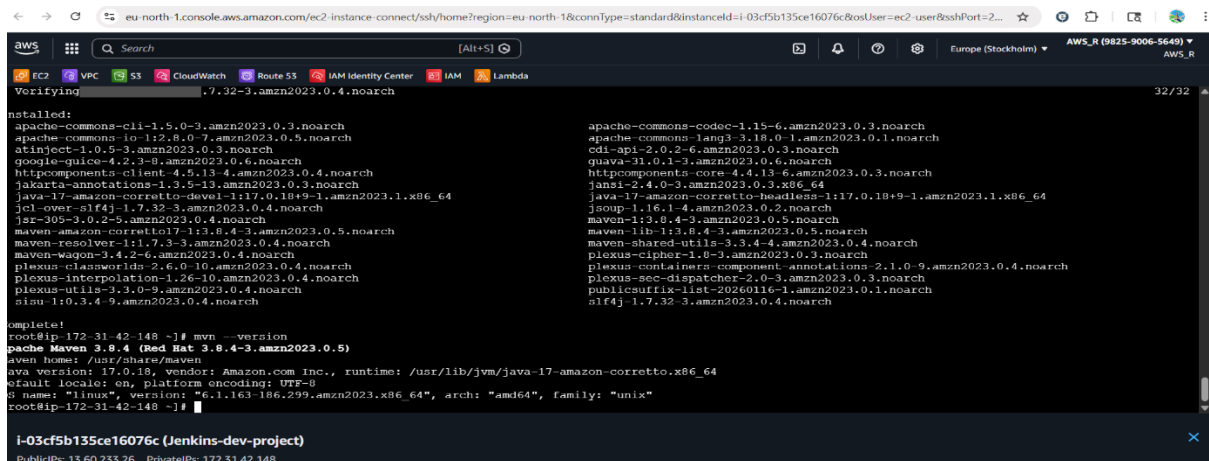
➔ Jenkins is ready to use



➔ Jenkins -> manage Jenkins ->System



➔ Jenkins -> manage Jenkins ->Tools



← → ↻ Not secure 13.60.233.26:8080/manage/configureTools/ ☆ ⓘ ⌵ 🔍 🌐

Jenkins / Manage Jenkins / Tools 🔍 ⚙️ 👤

Maven installations ^ ✎ Edited

+ Add Maven

⋮ Maven ✖

Name

MAVEN_HOME

MAVEN_HOME

/usr/share/maven

☐ Install automatically ?

+ Add Maven

Save Apply

* Now Jenkins is ready to work on.

TASK-4

Webhook Trigger =>

This is the most commonly used trigger in modern DevOps pipelines.

How it works

- 1)A webhook from the repository notifies Jenkins instantly.
- 2)No polling required.

Flow

-> Developer push → GitHub webhook → Jenkins → Build triggered

- 1)Name -> Web_Hook
- 2)Source code management

Jenkins / Web_Hook / Configure 🔍 ⚙️ 👤

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-Build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/rajeshtan114y/onlinebookstore-ashu.git

Credentials ?

none

Advanced ▾

+ Add Repository

Branches to build ?

Save Apply

3)Triggers

Jenkins / Web_Hook / Configure 🔍 ⚙️ 👤

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-Build Actions

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

☐ Trigger builds remotely (e.g., from scripts) ?

☐ Build after other projects are built ?

☐ Build periodically ?

☒ Call site hook trigger for GitHub polling ?

☐ Poll SCM ?

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

☐ Delete workspace before build starts

☐ Use secret text or file(s) ?

☐ Send files or execute commands over SSH before the build starts ?

☐ Send files or execute commands over SSH after the build runs ?

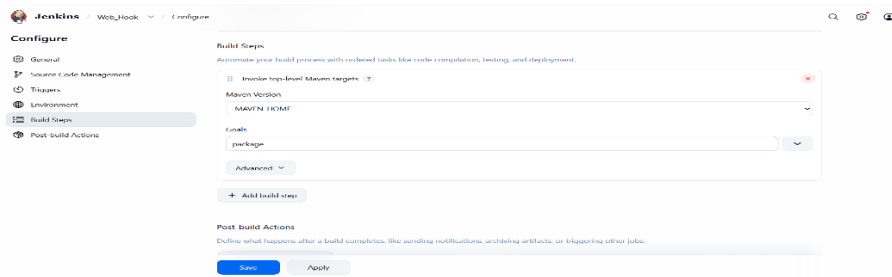
☐ Add timestamps to the Console Output

☐ Inspect build log for published build scans

☐ Terminate a build if it's stuck

Save Apply

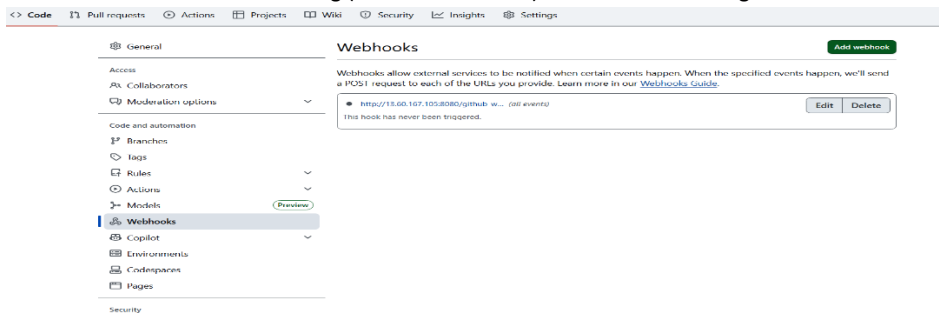
4)Build Step's



Then click on apply and Save.

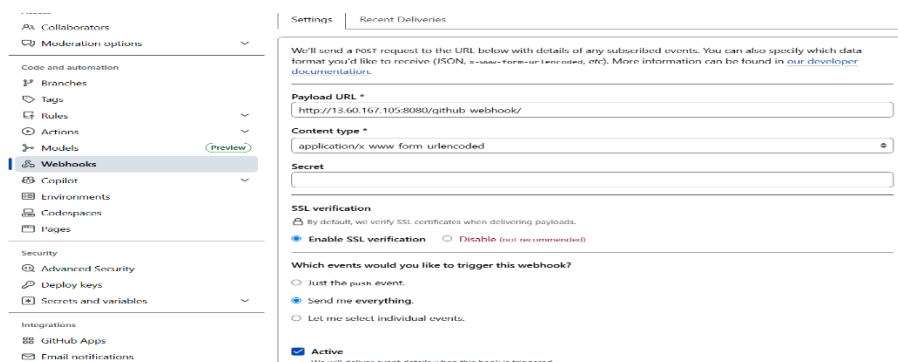
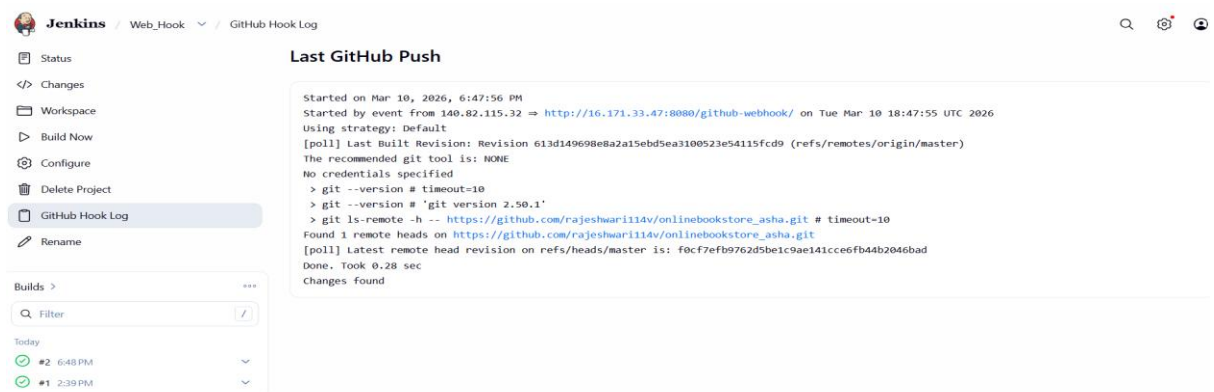
TASK-3

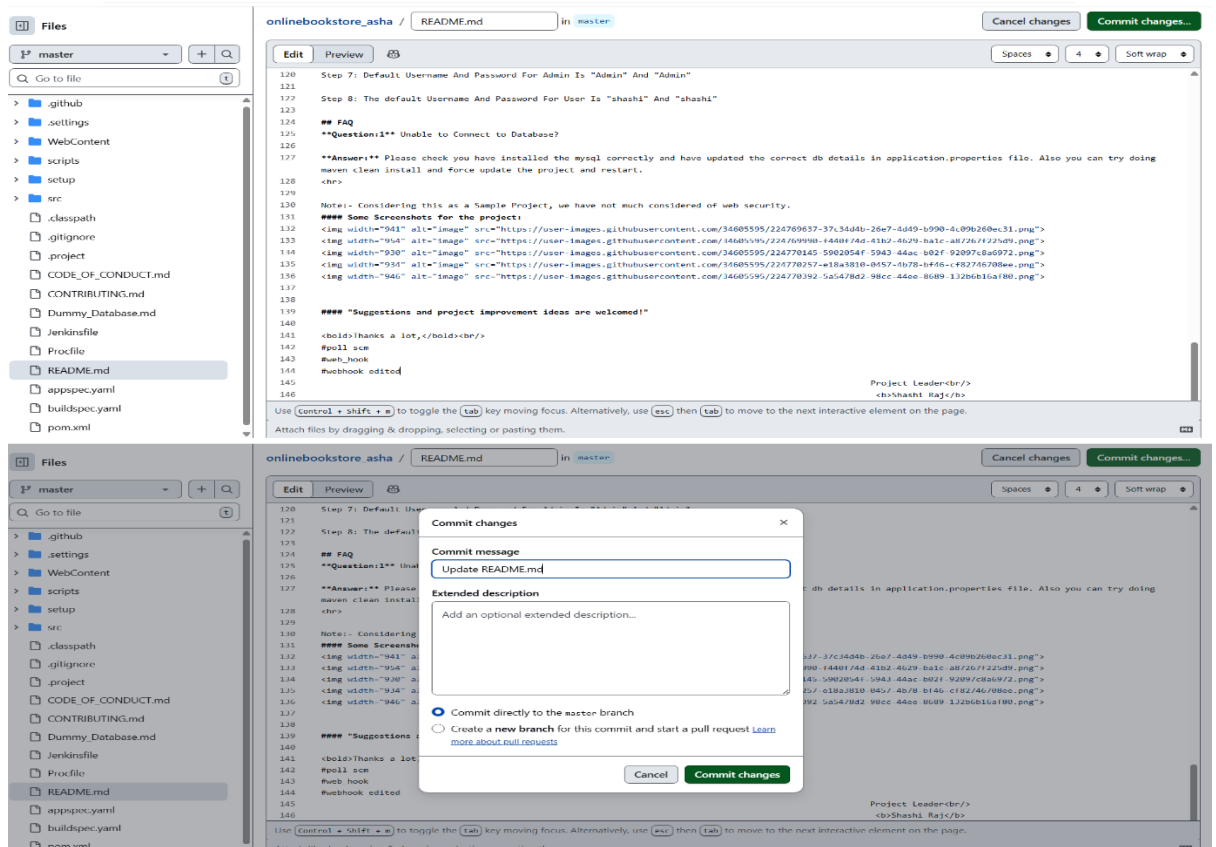
➔ Then click on GitHub Webhook log (left side of the tab) do the bellow setting's



Go to GitHub-> settings

1)click on Webhooks-> Add webhook -> Give confirm Access -> click on Add webhooks



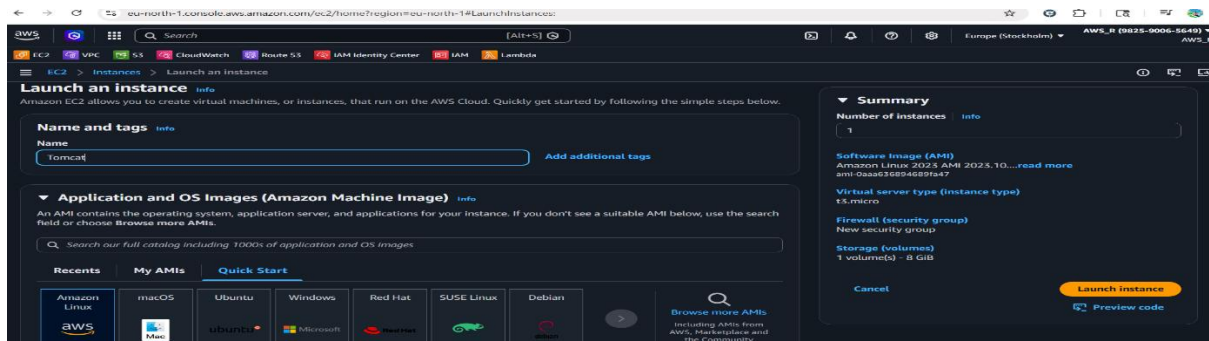


Then click on commit changes and reload the Jenkin page that will be triggered and show the result.

TASK-5

Apache Tomcat Setup & SSH Deployment • Install Apache Tomcat (version 9 or 10) on the target server. • Configure a Tomcat manager user with role manager-script in tomcat-users.xml. • Ensure Tomcat is running on port 8080 (or configure a non-conflicting port). • Configure SSH-based deployment from Jenkins to Tomcat server: ◦ Generate SSH key pair on the Jenkins server. ◦ Copy the public key to the Tomcat server's authorized keys.

➔ launch instance Apache Tomcat server and do the setting's below shown:



Search [Alt+S]

EC2 VPC S3 CloudWatch Route 53 IAM Identity Center IAM Lambda

EC2 > Instances > Launch an instance

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type	Protocol	Port range
ssh	TCP	22

Source type: Anywhere

Source: 0.0.0.0/0

Description - optional: e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 8080, 0.0.0.0/0)

Type	Protocol	Port range
Custom TCP	TCP	8080

Source type: Custom

Source: 0.0.0.0/0

Description - optional: e.g. SSH for admin desktop

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023 AMI 2023.10...read more

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 8 GiB

Cancel Launch instance Preview code

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:

EC2 > Instances > Launch an instance

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Add security group rule

Configure storage

1x 30 GiB gp3 Root volume, 5000 IOPS, Not encrypted

Add new volume

Click refresh to view backup information. The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

Advanced details

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023 AMI 2023.10...read more

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 30 GiB

Cancel Launch instance Preview code

➔ Then connect the Tomcat server.

```

aws
Search [Alt+S]
EC2 VPC S3 CloudWatch Route 53 IAM Identity Center IAM Lambda
EC2 > Instances > Launch an instance
HostKey /etc/ssh/ssh_host_ed25519_key
HostKey /etc/ssh/ssh_host_ed25519_key
#Cipher and Keying
#KeyLimit default none
#Logging
SyslogFacility AUTH
LogLevel INFO
#Authentication:
LoginGraceTime 2m
PermitRootLogin yes
StrictModes yes
MaxAuthTries 6
MaxSessions 10
PubkeyAuthentication yes
The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile .ssh/authorized_keys
AuthorizedPrincipalsFile none
- INSERT --
i-00e20d034e744575b (Tomcat)

```

```

aws
Search [Alt+S]
EC2 VPC S3 CloudWatch Route 53 IAM Identity Center IAM Lambda
EC2 > Instances > Launch an instance
MaxAuthTries 6
MaxSessions 10
PubkeyAuthentication yes
The default is to check both .ssh/authorized keys and .ssh/authorized keys2
but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile .ssh/authorized_keys
AuthorizedPrincipalsFile none
For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
HostbasedAuthentication no
Change to yes if you don't trust ~/.ssh/known_hosts for
HostbasedAuthentication
IgnoreUserKnownHosts no
Don't read the user's ~/.rhosts and ~/.shosts files
IgnoreRhosts yes
Explicitly disable PasswordAuthentication. By presetting it, we
avoid the cloud-init set_passwords module modifying sshd config and
restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no
Change to no to disable s/key passwords
- INSERT --
i-00e20d034e744575b (Tomcat)

```

← → 🔍 Not secure 16.171.15.247:8080/manage/ ☆ 🔄 📄 📄 📄 📄 📄 📄

Jenkins / Manage Jenkins 🔍 ⚙️ 👤

Manage Jenkins

🔍 Search settings /

Building on the built-in node can be a security issue. You should set up distributed builds. See the [documentation](#). [Set up agent](#) [Set up cloud](#) [Dismiss](#)

Jenkins can enforce Content Security Policy (CSP). CSP tells web browsers what they are allowed to do while rendering a web page. This limits or even eliminates the impact of vulnerabilities like cross-site scripting (XSS). CSP is disabled by default for backward compatibility, but it is recommended to enable it, if possible. [Learn more](#) [Dismiss](#)

System Configuration

- **System**
Configure global settings and paths.
- **Tools**
Configure tools, their locations and automatic installers.
- **Plugins**
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
- **Nodes**
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
- **Clouds**
Add, remove, and configure cloud instances to provision agents on-demand.
- **Appearance**
Configure the look and feel of Jenkins

```
[root@ip-172-31-19-151 ~]# cd
[root@ip-172-31-19-151 ~]# passwd root
Changing password for user root.
New password:
BAD PASSWORD: The password is a palindrome
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-19-151 ~]# vi /etc/ssh/sshd_config
[root@ip-172-31-19-151 ~]# service sshd restart
Redirecting to /bin/systemctl restart sshd.service
[root@ip-172-31-19-151 ~]# chmod 777 /opt/
[root@ip-172-31-19-151 ~]#
```

← → 🔍 Not secure 13.48.195.228:8080/manage/configure ☆ 🔄 📄 📄 📄 📄 📄 📄

Jenkins / Manage Jenkins / System 🔍 ⚙️ 👤

name

tomcat

Hostname ?
172.31.19.151

Username ?
root

Remote Directory ?
/opt

☐ Avoid sending files that have not changed ?

Advanced ▾

Test Configuration

← → 🔍 Not secure 13.48.195.228:8080/manage/configure 🗄️ ☆ 🔄 📄 📄 📄 📄 📄 📄

Jenkins / Manage Jenkins / System 🔍 ⚙️ 👤

Remote Directory ?
/opt

☐ Avoid sending files that have not changed ?

Advanced ▴  Edited

☒ Use password authentication, or use a different key ?

Passphrase / Password ?
-

Path to key ?

Key ?